



Interfaces with Other Disciplines

Assessing the impact of jumps in an option pricing model: A gradient estimation approach

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ABSTRACT

Motivated by model risk considerations, we develop a statistical procedure that determines whether the inclusion of a jump component in a simpler, diffusion-based price model significantly influences the prices of specific options on this underlying.

The basis of our statistical testing procedure is simulating the sensitivity of the option price in the framework of jump-diffusion Markov Chains. The jumps are assumed to follow a compound Poisson process. The stochastic gradient representation of the resulting model risk is general: it can be applied to quantify the difference between performance functions of two Markov Chains at multiple future times. The sensitivity estimator samples the jump-diffusion within the base diffusion process.

We show that our statistical test is vastly superior to the two-sample *t*-test. We also demonstrate that the test is particularly powerful in situations where either the volatility or the jump component is dominant.

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1. Motivation

1.1. Model risk

"All models are wrong but some are useful", as a renowned British statistician George Box famously said. This is particularly true in finance and especially in option pricing: the most widely used option pricing model is still the famous Black and Scholes model, almost 50 years after its introduction. Despite its several shortcomings (e.g., asset returns are not normal, volatility is not constant), its popularity is so persistent exactly because it is so useful. Option traders like models that have closed-form (or nearly closed form) solutions, due to the required speed with which they must calculate option prices and greeks. More complex models might be more accurate in describing the dynamics of the underlying, but they may require solutions that are computationally involved and hence slow, such as Monte Carlo simulations. Furthermore, more complex models usually also require cumbersome calibration techniques that can severely undermine this sought after accuracy. Hence, in practice, there is a constant search for

balance: between accuracy and speed, and between simplicity and complexity.

One classical example of this struggle is the question of whether prices of assets underlying option contracts exhibit jumps. A more practical question, however, would be: does it matter for option prices? If including jumps does not significantly influence option prices, then a simpler, diffusion-based model is preferred in practice. This is especially true as more complex models (such as jump diffusions) have a substantial model risk associated with them, and model risk is currently one of the main points of attention for regulators and financial institutions.

Existing methods to assess the presence of jumps in the underlying asset prices is to apply some statistical analysis to observed price paths (Aït-Sahalia, 2004; Ball & Torous, 1985, etc). However, it is well-known that most of these statistical methods overestimate the jump frequency. One asset where jumps are definitely present in prices is electricity (e.g., Cartea & Figueroa, 2005; Erlwein, Benth, & Mamon, 2010; Mayer, Schmid, & Weber, 2015). In these papers, statistical methods for estimating jump frequency are applied to electricity prices, and it is found that approximately 40% of price moves can be classified as jumps. However, from our practical experience in energy trading, we observe that electricity traders and analysts put this percentage in the range of 5–10%. In commodities other than electricity (and other asset classes such

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as FX) it is not even clear whether jumps are present in price trajectories, or whether a diffusion-based price model would suffice. However, many authors do include jumps into e.g., FX or oil price models; see e.g., Hilliard & Reis (1999), Clewlow & Strickland (2003), Larsson & Nossman (2011). Furthermore, pure jump processes such as Levy processes are also used in academic literature to model asset prices. All these finite or infinite activity jump processes significantly increase the complexity of underlying models.

In practice, financial institutions prefer to avoid additional model complexity whenever possible, as increased complexity leads to higher model uncertainty and model risk.

Jump diffusions are considered particularly prone to model risk (more so than e.g., stochastic volatility models), as calibration of these models is typically cumbersome, difficult, and prone to errors. These difficulties are well-documented, see e.g., Cont and Tankov (2004b).

So from the practical perspective, a model for the underlying that does not contain jumps is preferred, at least if the difference in option pricing outcomes is not significant. Another practical consideration is that simpler models lead to lower model development and maintenance costs as well as less effort in model validation a consideration which is particularly relevant in the current environment when banks are forced to significantly cut costs. Finally, increasing regulatory pressure also contributes to banks' attitude towards complex models: recent regulation such as TRIM (Targeted Review of Internal Models, Guide for the Targeted Review of Internal Models TRIM, 2017) puts an increasing focus on understanding, measuring, and managing model risk in financial institutions.

In option pricing, model risk is often measured by the difference in option prices between different pricing models, or between the model-based and quoted option prices. Derman (1996) defines model risk in derivatives pricing as “the risk that the model is not a realistic or plausible representation of the factors affecting the derivative's value”. On the other hand, Morini (2011) defines it as ‘the risk of a significant difference between the mark-to-model value of an instrument, and the price at which the same instrument is revealed to have traded in the market’.

Academic literature on measuring and mitigating model risk in finance is still quite sparse and is lagging behind the regulatory efforts. Quantifying model risk in a coherent framework is proposed and studied in works by Cont & Hamida (2005), Cont (2006), Cont & Tankov (2006). Glasserman & Xu (2014) use the concept of relative entropy to quantify model risk. Detering & Packham (2016) quantify potential losses arising from using misspecified models when pricing and hedging contingent claims. Cont (2006) address model risk in option pricing by two different methods. The first method, suitable for a single option, considers the maximal difference in option prices within a class of considered pricing models and measures the model risk as this difference relative to the total portfolio position. The second method determines the best pricing model, by minimizing the calibration error with regard to a collection of traded options. The authors cast these two procedures in the framework of the existing risk measures. They observe that model risk is option-dependent. Cont & Tankov (2004a) choose the “best” model, by minimizing the relative entropy of candidate models.

The approach we take in this paper close in spirit to Derman's definition of model risk. Our proposed methodology helps in deciding whether including jumps in the underlying dynamics has a significant effect on option prices. We do not debate whether there are or there aren't jumps in asset prices; instead, we address this issue from the model risk perspective: do we significantly underestimate the value of the option by considering a simpler, diffusion-based model rather than a more complex, jump-diffusion model? Towards this end, we develop a sensitive and powerful statisti-

cal testing procedure, which utilizes price paths that can occur in the competing models. So our statistically sound new methodology helps in discerning whether the marginal improvement in pricing accuracy/quality justifies increased model complexity and hence model risk.

It is worth mentioning that ours is not the first paper taking this approach. For example, Benth & Schmeck (2014) address the same issue in the context of energy futures options and show that a simpler, Black'76 model often suffices for pricing futures options, even when energy spot price exhibits jumps.

1.2. Main methodological ideas

The main idea behind our methodology is the transition kernel representation of the stochastic price process. Typically, in hypothesis testing applications, the transition kernel is associated with the null hypothesis. Instead, here we consider a convex combination of transition kernels, associated with the null alternative hypotheses. Stochastic gradient estimation is then used to obtain a simulation-based estimator, by partitioning the transition kernel over the time to maturity of the option, according to some predetermined stopping times. This leads to the statistical testing procedure to discriminate between competing models. A typical example of a pair of such competing models is the geometric Brownian motion, in short, GBM (null hypothesis) and the Merton Model, i.e., the GBM with a compound Poisson jump process (alternative hypothesis). For this pair of models, the stopping times are the simulated jump times from the Merton model as well as the start and end of the option contract. The power of our test can be interpreted as the proportion of the alternative model in the mixed transition kernel.

A second, supplementary, contribution of this paper is the new proposed structure of the stochastic gradient estimator. We implement the gradient-based technique in the simulation estimator via the so-called *Measure Valued Differentiation (MVD)* (Heidergott & Volk-Makarewicz, 2015; Pflug, 1996). The MVD differentiation approach yields a difference of two stochastic processes: the process under the null hypothesis and the process with a non-homogeneous transition kernel. Under our consideration of Markov processes, this gradient price path is composed of “gluing” the paths simulated from the alternative hypotheses to the paths from the null hypothesis. In the context of jump diffusions, we determine two successive stopping times and replace GBM with the diffusive component of the Merton Model between these times. If the first stopping time is a jump time, we then include a jump into the price process. The use of stopping times provides variance reduction and simulation efficiency in determining sensitivities of stochastic price paths. The approach of this paper has connections to both multilevel simulation, e.g. Giles (2008), and adjoint differentiation techniques, e.g. Capriotti & Giles (2011).

The paper is structured as follows. In Section 2, we outline the statistical testing framework. In Section 3, we provide sufficient conditions for the analyticity of the option price with respect to an artificial mixing parameter, and the results provided in this section are the mathematical foundation of our new gradient estimators. A numerical study is provided in Section 4. The last section concludes and provides directions for further research.

2. Statistical testing framework

2.1. Preliminaries

Before discussing the problem, we first describe the tools that we use. Let $X = (X(t))$, $Y = (Y(t))$, $t \in [0, T]$ be two strongly Markovian stochastic processes, that is, given a jump time $\tau \in [0, T]$, $t > \tau$ it holds that $X_t - X_\tau = X_{t-\tau}$ with probability one. With

each stochastic process, for $t \in [0, T]$, there is an associated transition functions P^t for X and Q^t for Y . The stochastic processes possess a common initial value $X(0) = Y(0) = x_0$. The stochastic processes exist within a (filtered) common Borel probability space $(\Omega, \mathcal{F}, (\mathcal{F})_t, \mathbb{Q})$ that satisfies the usual conditions.

Define observation times as dates when an option contract is observed, other than maturity.¹ For the stochastic process X , we denote the values at observation times $t = (t_0, \dots, t_o)$, with $0 < t_1 < \dots < t_o := T$ by $u = (u_1, \dots, u_o)$.

We denote the value of a European option on our stochastic process X at maturity T by

$$\mathbb{E}_{u_0}[\Phi(X(T))] := P_{x_0}^T \Phi := \int_{\mathbb{R}^n} P^T(du_0; u_0) \Phi(x) \quad (1)$$

where $\Phi: \mathbb{R}^o \rightarrow \mathbb{R}$ denotes the payoff of an contingent claim. By partitioning on the observation times, we note that by the semi-norm property of Markov processes we have

$$P^T(dx; x_0) := \prod_{j=1}^o P^{t_j - t_{j-1}}(dx_j; x_{j-1}). \quad (2)$$

The market option price is denoted by $O_{x_0}^T \Phi$. Since we are considering option valuation, we are working under the Equivalent Martingale Measure $\mathbb{Q}$.

We require our option payoff functions to be L^2 integrable. The requirement of the strong Markov property is not onerous, as this includes solutions of stochastic differential equations and integrals of Lévy processes, so almost all common models used in quantitative finance.

Example 1. Let r be a deterministic risk-free interest rate, let $\sigma > 0$ be the volatility parameter (assumed constant) and $W = (W_t), 0 \leq t \leq T$ is the Wiener process. Considering logarithmic price process $X_t = \ln S_t$, under the EMM this process has the following dynamic:

$$dX_t = (r - \sigma^2/2) dt + \sigma dW_t. \quad (3)$$

Denoting

$$\mu := (r - \sigma^2/2), \quad (4)$$

with initial price S_0 , the price process can be solved to be the stochastic exponential $S_t = S_0 \exp(\mu t + \sigma W_t)$. For fixed t , the marginal of the stochastic process is normally distributed $X_t \sim \mathcal{N}(\mu t, \sigma^2 t)$ with mean μt and variance $\sigma^2 t$. Since the Wiener process has identically distributed increments, for times $s < t$ the transition density from $X_s = x$ to X_t is only dependent on the lag

$$p^u(dy, x) = \frac{1}{\sqrt{2\pi\sigma\sqrt{u}}} \exp\left(-\frac{1}{2} \frac{(dy - x - \mu u)^2}{\sigma^2 u}\right).$$

Due to the Strong Markov property, see [Rogers & Williams \(2000\)](#), between stopping times τ, η , for $0 \leq \eta < \tau \leq T$, the random variable $X(\tau)$, given $X(\eta) = x$, can be generated via a standard $Z \sim \mathcal{N}(0, 1)$ normal random variable where

$$X(\tau) = x + \mu(\tau - \eta) + \sigma\sqrt{\tau - \eta}Z$$

and, for $t > s > 0$, under P^{t-s} ,

$$S(t) = S(s) \exp(\mu_D(t - s) + \sigma\sqrt{t - s}Z).$$

The Merton model extends the GBM through the inclusion of (usually normally distributed) jumps, where the time between consecutive jumps is exponentially distributed. In between, the stochastic process evolves just like a GBM. For the jump size J we

assume a lognormal distribution, more specifically, we assume that $\ln(1 + J) \sim N(\kappa, \delta)$. The random variable J is independent of both W and the Poisson process N_t . Let Y_t be the log-price under the Merton model, and, let $\gamma = \lambda \bar{J}$, where $\bar{J} = \mathbb{E}[J] = e^{\kappa + \delta^2/2} - 1$, with λ being the intensity of the Poisson process. From [Cont & Tankov \(2006\)](#), the stochastic differential equation under the EMM is

$$d \ln Y_t = (r - \sigma^2/2 - \gamma) dt + \sigma dW_t + J dN_t. \quad (5)$$

The term γ is to ensure that the compound Poisson process is a martingale under $\mathbb{Q}$ and Y_{t-} is the value of price process prior to the possible inclusion of a jump at time t . Let

$$\mu_J := \mu_D - \gamma. \quad (6)$$

Since the value of the Poisson process is only dependent on the lag we obtain the transition density from $Y_s = x$ to Y_t by

$$Q^u(dy, x) = e^{-\lambda u} \sum_{m=0}^{\infty} \frac{(\lambda u)^m \exp\left(-\frac{1}{2} \frac{(dy - x - \mu_J u - m\kappa)^2}{\sigma^2 u + m\delta^2}\right)}{m! \sqrt{2\pi(\sigma^2 u + m\delta^2)}}, u = t - s,$$

where the infinite sum stems from the fact that $q_{t,s}$ has to take into account all possible number of jumps.

When constructing the GBM and the Merton model on a joint probability space, one can use the following maximal coupling scheme. Let the GBM have value $X(\tau_m) = x$ and let under the Merton Model the value by y , then the value of the Merton Model at time of the next jump τ_{m+1} satisfies

$$\ln Y_{\tau_{m+1}} = \ln y + (\ln X_{\tau_{m+1}} - \ln x) + J_{m+1} - \gamma(\tau_{m+1} - \tau_m)$$

and the marginal at time jump time τ_{m+1} is distributed as

$$S(\tau_{m+1}) = J_{m+1} + S(\tau_m) \exp(\mu_J(\tau_{m+1} - \tau_m) + \sigma\sqrt{\tau_{m+1} - \tau_m}Z).$$

In words, in-between jumps the GBM and the Merton Model follow the same probabilistic dynamic driven by the Wiener process. It is, however, worth noting that the risk-free interest rate differs by the jump rate γ , i.e., $\mu_D = \mu_J + \gamma$.

Example 2. Let $W = (W_t), t > 0$ be the Wiener process. Let κ and μ be parameters where κ denotes the rate of reversion to the long term mean μ . Defining the price process as $S_t = \tilde{X}_t$ with initial price s_0 , the arithmetic mean-reverting model (also called Vasicek model) is defined under the real-world measure as

$$d\tilde{X}_t = \kappa(\mu - \tilde{X}_t) + \sigma dW_t. \quad (7)$$

The benefit of the arithmetic form of the price process is that the process X can take negative values. This makes this process excellently suited to modeling e.g., short interest rate (which has been negative in Europe for the past two years) or electricity prices. Because electricity is non-storable, negative prices are observed in electricity markets during occasions of extreme oversupply. Such negative prices also have been recently observed in oil markets, due to huge oversupply in the midst of COVID-19 pandemic and the subsequent lack of available storage. Negative prices mean that market participants must be compensated for taking over the excess electricity (or crude oil). Note that we also consider an exponential form of mean reversion (the so-called Exponential Vasicek model), in which case all the derivations work similarly.

The Mean Reversion Jump-Diffusion (MRJD) model, see [Clewlow & Strickland \(2003\)](#), [Borovkova & Schmeck \(2017\)](#), as specified by its name, is a price process that combines a mean reversion diffusion component with a compound Poisson process.

Let $N = (N_t)$ be a Poisson Process with rate λ and let J be the jump size. We can again use normal distribution, but in our application of this model we will use an asymmetric double exponential distribution. It is specified by the corresponding parameters $\lambda^\pm > 0$. If we denote the probabilities of jump in the positive and

¹ For example, these could be the observation times for Asian options, or, for barrier options, the observation times when the asset value is checked against the barrier, to determine whether the option contract is cancelled or initiated.

Table 1
Contract and model parameters for Model Pair 1.

Parameter settings for Model Pair 1	
Contract Parameters	$s(0) = 40, K = 40, T = 1.$
Model Parameters	
H_0 : GBM	$r = 0, \sigma = 1.698 \times 10^{-2}.$
H_1 : Merton Model	$r = 0, \sigma = 1.698 \times 10^{-2}, \lambda = \{0.01, 0.08, 0.70\}, \mu_J = -3.3 \times 10^{-5}, \delta = 3.347 \times 10^{-2}.$

Table 2
Contract and model parameters for Model Pair 2.

Parameter settings for Model Pair 2	
Contract Parameters	$s_0 = 40, K = 35, T = 1$
Model Parameters	
H_0 : Vasicek Model	$r = 0, \kappa = 0.24, \mu = 0.63, \sigma = 5.16.$
H_1 : MRJD Model	$r = 0, \kappa = 0.24, \mu = 0.63, \sigma = 5.16, \lambda = \{0.01, 0.08, 0.70\}, \beta^+ = 3.48, \beta^- = 5.54$

negative direction by $\pi^\pm$, with $\pi^+ + \pi^- = 1$, the double exponential density function is given by

$$f_J(x) = \pi^+ \lambda^+ e^{-\lambda^+ x} \mathbf{1}\{x > 0\} + \pi^- e^{-\lambda^- |x|} \mathbf{1}\{x \leq 0\}.$$

If $E_1(\lambda^+), E_2(\lambda^-)$ are independent exponentially distributed random variables with parameters λ^+, λ^- respectively, and U is a uniform random variable on $(0,1)$, the jump J can be simulated via $J = \mathbf{1}\{U \leq \pi^+\} E_1(\lambda^+) - \mathbf{1}\{U > \pi^+\} E_2(\lambda^-)$. The inclusion of the jump component yields the expression for the MRJD Model under EMM:

$$dY(t^-) = \kappa \left(\mu - \lambda^M \sigma - \frac{\lambda \mathbb{E}[J]}{\kappa} - Y(t^-) \right) dt + \sigma dW(t) + J dN(t). \quad (8)$$

2.2. To jump or not to jump: practical considerations

When choosing an appropriate option pricing model, we need to balance two considerations: using a model with additional features (that can lead to a more “accurate” description of the underlying price process) and adding accuracy in pricing our options. Additional features (such as jumps) can more accurately describe the evolution of the underlying, but they do not always contribute towards “better” option prices (e.g., prices that are closer to those quoted in the market or those that better match hedging costs). This can be a result of several things. For instance, in Table 2 of Bates (1996), although there is a benefit in incorporating Merton model jumps in a diffusion and a SV model, part of this benefit arises from the reduction of volatility when jumps are included into the CIR and Heston models.

The effect of including jumps on option prices also depends on the type of option and its characteristics. For example, Asian options are less affected by jumps due to averaging, as are options with long maturities. So it is a combination of the underlying process and the option in question (and not just the characteristics of the underlying) that determines the decision whether a more or less complex model should be used. This decision should preferably be based on a quantitative assessment of economic significance (on option prices) of including jumps.

It is well-known that, with jump-diffusions we are better able to calibrate the dynamics of the volatility smiles (see e.g., Andersen & Andreasen, 2000). However, if the goal is to value particular options (rather than model implied volatilities), this application of jump-diffusions becomes less relevant. In many cases, option prices can be, in theory, similar for the competing models (basic and more advanced), but the potential for calibration error and model misspecification is much higher for complex models.

To illustrate that, consider a situation similar to that in the original Merton's paper. We price a simple call with 6-month ma-

turity, $S_0 = 38, K = 35, r = 0.1$ and variance of the diffusion part $\sigma^2 = 0.05$. Assume that there is one jump per year ($\lambda = 1$) and that the jump size is lognormally distributed with parameters μ and δ^2 . Let $\delta^2 = 0.05$ - and moreover, assume that we “know” the two jump parameters λ and δ^2 , or at least are confident in their calibrated value. This leaves one more jump parameter to be calibrated, and that is μ . This parameter is particularly hard to calibrate (as it is the case with all first-moment-like quantities), so statistical or calibration error for its estimate can be very large. Assume that the true value of this parameter is -0.25 . Merton's option price in this case is 6.69. The Black-Scholes price, computed with the historical estimated total volatility of this jump-diffusion process will be 6.81, so not that far away from the Merton's price. On the other hand, if the estimate of μ is, say, -0.13 (which is not unrealistic, given potentially high confidence interval around this estimate), then the Black-Scholes price of course remains the same (6.81) but Merton's option price becomes 6.21 - meaning that the option is now almost 9% underpriced. Often the parameter μ is not even estimated but simply assumed to be zero. In this case, the Merton price becomes 5.97 - almost 13% underpricing the option. This shows how additional model complexity (i.e., including jumps) and the resulting uncertainty in parameter estimates can severely affect option prices.

The one seemingly obvious way to decide upon the appropriate model is to include jumps (if there is any suspicion that those could be present in the price process), fit the jump-diffusion - like model to the data, and then decide on the basis of whether the jump parameters (frequency and severity) are significantly different from zero. Note, however, that in such a situation only the underlying price process is used to make the model choice and not the option(s) that must be valued. In many situations it can happen that, even when the estimated jump parameters are significantly different from zero, including the jump component into the model does not lead to a significant effect on option prices, precisely due to one of the points mentioned above: a longer maturity of an option or its type such as Asian feature. In these cases, a simpler model is still preferred, due to the model risk and model uncertainty considerations outlined in the introduction. Furthermore, reliably calibrating jump parameters is quite difficult in practice (and these difficulties are well-documented, see e.g., Cont & Tankov, 2006), and as we saw above, errors in calibration can lead to large errors in option prices.

2.3. An alternative: t-test

Another way to discern between competing option pricing models (or the same model but with two different calibration outcomes) is by means of a simple t-test. Let the incumbent option

pricing model be given by the transition function P^t (the X process) and alternative model is the model with the transition function Q^t (the Y process). The model associated with the market observed option value (the V process) as transition function O^t . Recall that option values are expectations of discounted payoffs, which can be easily computed by simulations, replacing the expectation by the average payoff. In such stochastic simulation, there is an estimation error, due to estimating the expectation by the sample mean. Suppose that simulations of the payoffs have been obtained with n i.i.d. underlying price paths. Let $\widehat{P_{x_0}^T \Phi}$ and $\widehat{Q_{x_0}^T \Phi}$ be the simulated option prices under both models. We consider the two-sided hypothesis test in which the null hypothesis is that there is no significant change in option value relative to the market price, against the alternative hypothesis that the deviation to the market option values are different, i.e.,

$$\begin{aligned} H_0 : \widehat{Q_{x_0}^T \Phi} - O_{x_0}^T &= \widehat{P_{x_0}^T \Phi} - O_{x_0}^T, \\ H_1 : \widehat{Q_{x_0}^T \Phi} - O_{x_0}^T &\neq \widehat{P_{x_0}^T \Phi} - O_{x_0}^T. \end{aligned} \quad (9)$$

Because

$$(\widehat{Q_{x_0}^T \Phi} - O_{x_0}^T) - (\widehat{P_{x_0}^T \Phi} - O_{x_0}^T) = \widehat{Q_{x_0}^T \Phi} - \widehat{P_{x_0}^T \Phi}, \quad (10)$$

the test statistic is formulated from the difference in computed values between the competing models. Denote by $\alpha \in (0, 1)$ the type I error. The form of the test statistic depends upon the method of simulation used for $\widehat{P_{x_0}^T \Phi}$ and $\widehat{Q_{x_0}^T \Phi}$. If Common Random Numbers² (CRN), Asmussen & Glynn (2007), are used in computing of $\widehat{P_{x_0}^T \Phi}$ and $\widehat{Q_{x_0}^T \Phi}$, then we can define the variance of our test statistics to be the variance of the differences of the option prices at each path. Let s_{crn}^2 be this value. The standardized test statistics under the null hypothesis $T_{0,crn} \sim t_{n-1}$ is t-distributed with $n - 1$ degrees of freedom:

$$T_{0,crn} = \sqrt{n} \frac{\widehat{Q_{x_0}^T \Phi} - \widehat{P_{x_0}^T \Phi}}{s_{crn}} \sim t_{n-1}. \quad (11)$$

Rather than using CRN, option prices for the two models can be independently estimated. Let s_P^2, s_Q^2 be the variances in the option payoff under the respective models P^t, Q^t . The variance of the difference in option prices in this case is the sum of two variances: $s_{ind}^2 = s_P^2 + s_Q^2$. The test statistics $T_{0,ind}$ for this test is the same as in Eq. (11), with s_{ind} replacing s_{crn} . The random variable $T_{0,crn} \sim t_\nu$ where the computation for degrees of freedom ν can be found in any introductory textbook in statistics, for instance in Section 8.4 of Dowdy, Weardon, & Chilko (2005).

The rejection of the null hypothesis informs us that option price under model P is significantly different than under model Q . However, this does not necessarily inform us that price model Q is more accurate than P . The reason is that, given there is a difference in option price between the models, the null hypothesis will be always rejected for a sufficiently high number of price paths.

We build an alternative test by introducing an option price “tolerance”, i.e., we introduce a “significant” difference in the deviation of option prices to the market value. Let $t_{\beta,\gamma}$ be the γ -quantile, $\gamma \in (0, 1)$, of the t_β distribution, $\beta > 0$. Let $\delta = d(P^T \Phi - O^T \Phi)$ be the pre-specified deviation in option price difference between the two models considered to be significant for switching from P^t to Q^t . Then the test statistic under the alternative hypothesis, $T_{1,crn}$, is

$$T_{1,crn} = \sqrt{n} \frac{\widehat{Q_{x_0}^T \Phi} - \widehat{P_{x_0}^T \Phi} - \delta}{s_{crn}} \sim t_{n-1}. \quad (12)$$

² CRN is a variance reduction technique where the same underlying random numbers are used for simulating paths the performance of which are compared.

The test power at level α is then the probability $\mathbb{P}(|T_{1,crn}| > t_{n-1, 1-\alpha/2})$. This value can help us decide whether there is a significant (with respect to the tolerance level δ) difference in the deviation of option prices to the market value under the two models to justify using (a more complex) model Q over a simpler model P .

The cancellation of the observed market price (10) applies to any time interval. Consequently, only the difference of the competing prices are needed in the development of the proposed estimator,

2.4. New test statistics

The use of t-test for the option model choice setting is easy to understand and implement. However, it is less appropriate when we want to explicitly distinguish between models with and without jumps. This is because we have additional information about what kind of difference between the models we are testing for, and this is not reflected in a very general character of the t-test. In this section, we built an alternative statistical test that is specifically tailored to the alternative underlying process specification, namely a base diffusion process plus Poisson jumps.

Let $N := N(T)$ in which $N(t), t \in [0, T]$, be a Poisson process with rate $\lambda > 0$. We denote the jump times of $N(t)$ by τ_i , for $i = 0, \dots, N(T)$, with $\tau_0 = 0, \tau_{N(T)+1} = T$, and $\tau_i < \tau_{i+1}$, for $0 \leq N(T) - 1$. The jump size is denoted by J , and J_i is the jump size at τ_i . The compounded process of jumps has at time t value

$$J(t) = \sum_{i=0}^{N(T)} \mathbf{1}_{\tau_i \leq t} J_i, \quad 0 \leq t \leq T.$$

We assume that under the path measure Q^T there occur jumps of height J at jump times of Poisson process. To be precise, we assume that Y has cadlag paths so that a Poisson jump at time t gives $Y(t^-) \neq Y(t) = Y(t^+)$. Put differently,

$$Y(\tau_i^-) + J_i = Y(\tau_i^+) = Y(\tau_i), \quad i = 1, \dots, N(T). \quad (13)$$

The specific Poisson process model depends on the assumed model, such as the Merton model and the mean reversion jump-diffusion, see Example 1.

The basic principle of our approach is that on the intervals (τ_i, τ_{i+1}) the stochastic dynamics of P and Q are identical and can therefore be coupled in a simulation. In words, we can drive a simulation of Q by using the random dynamic of P between τ_i and τ_{i+1} , and we may add jumps as an exogenous process to the simulated P transitions. It is, however, important that the Q and P dynamic between jumps differs by the different risk-free interest rates.

For deriving our new test statistics, condition on $N(T)$ and consider the path measure between Poisson jump times. To simplify the notation, we suppress denoting the observation times and denote by $p_i(y, x)$ the path likelihood of model X between Poisson jump time τ_i and τ_{i+1} . In the same way, we let $q_i(y, x)$ denote the path likelihood of model Y between Poisson jump times τ_i and τ_{i+1} , where the jump at time τ_{i+1} is included in the dynamic, see (13).

Before presenting the technical details, we will motivate the new estimator. Rather than considering the difference between P and Q by following the incumbent processes on the entire time interval $[0, T]$, we develop a framework that allows to “locally” compare the processes X and Y . For $\{\tau_i : 0 \leq i \leq N(T)\}$, we have

$$\begin{aligned} \mathbb{E}_{x_0}[\Phi(X)|N(t), 0 \leq t \leq T] \\ = \int_{\mathbb{R}^{N(T)+1}} p_{N(T)}(u_{N(T)+1}; u_{N(T)}) p_{N(T)}(u_{N(T)}; u_{N(T)-1}) \cdots p_0(u_1; x_0) \Phi(u) du. \end{aligned} \quad (14)$$

For the Q process we use similar notation. This discretization above allows to combine the P with the Q dynamic in a piece-

wise way, and we may "gradually" insert jumps. Given the jump time partitioning of $[0, T]$ we consider per element of the partitioning the convex combination of P and Q on that interval, i.e., $p_{i,\theta} = (1 - \theta)p_i + \theta q_i$. We let

$$p_\theta^T(x_{N(T)+1}, x_{N(T)}, \dots, x_0) = \prod_{i=0}^{N(T)} p_{i,\theta}(x_{i+1}, x_i)$$

and have the obvious relation that p_0^T recovers X and p_1^T recovers Y . As we will show in Section 3, under appropriate conditions, p_θ^T is analytical in θ and we can write

$$p_\theta^T = \sum_{n=0}^{\infty} \frac{\theta^n}{n!} (p_0^T)^{(n)},$$

where

$$(p_0^T)^{(n)} = \left. \frac{d^n}{d\theta^n} p_\theta^T \right|_{\theta=0}$$

denotes the n th order derivative of p_θ^T with respect to θ evaluated at $\theta = 0$. As we will show in Section 3, conditioned on the jumps process, it holds that

$$\begin{aligned} \left. \frac{d}{d\theta} \{p_\theta^T \Phi\} \right|_{\theta=0} &:= \left. \frac{d}{d\theta} p_\theta^T \right|_{\theta=0} (x_{N(T)+1}, x_{N(T)}, \dots, x_0) \\ &\quad \Phi(x_{N(T)+1}, x_{N(T)}, \dots, x_0) \\ &= \sum_{j=0}^{N(T)} \prod_{i=j+1}^{N(T)} p_i(x_{i+1}, x_i) (q_j(x_{j+1}, x_j) - p_j(x_{j+1}, x_j)) \prod_{i=0}^{j-1} p_i(x_{i+1}, x_i) \\ &\quad \Phi(x_{N(T)+1}, x_{N(T)}, \dots, x_0). \end{aligned} \quad (15)$$

If the above derivative is close to zero, it implies that adding jumps to the model (i.e., moving away from $\theta = 0$) has no immediate effect on the price as a mapping of θ , which may indicate that the jump component is not driving any noticeable price difference in the P and the Q model. This leads to the test statistic

$$G_0 = \sqrt{n} \frac{\left. \frac{d}{d\theta} \{p_\theta^T \Phi\} \right|_{\theta=0}}{\left(\left. \frac{d}{d\theta} \{p_\theta^T \Phi\} \right|_{\theta=0} \right)}, \quad (16)$$

where $s(\left. \frac{d}{d\theta} \{p_\theta^T \Phi\} \right|_{\theta=0})$ is the sample error of the simulated gradient estimator in (16) (for simulating of this gradient we will use CRN). In terms of hypothesis testing, the assertion $p_{x_0}^T \Phi = q_{x_0}^T \Phi$ implies $G_0 = 0$. We can then directly write a hypothesis test as

$$\begin{aligned} \text{first-order test} \quad H_0: G_0 &= 0 \\ H_1: G_0 &\neq 0. \end{aligned} \quad (17)$$

The test based on G_0 is testing whether having even a small degree of jumps has an impact on the price. If the price is very sensitive with respect to jumps, it is conceivable that $G_0 \neq 0$ will pick up the strong effect of jumps on the price. However, if the jumps have a more subtle effect, we will need a more elaborate test statistic. The basic idea of our approach is that if jumps have a strong explanatory power for the price difference, the effect on the price will predominately be driven by successive jumps. Indeed, given a reasonable level of volatility, the effect of a jump at time, say, t on the price at time $t + h$ may have been mellowed through the volatility of the price process on $[t, t + h]$. In the following, we will introduce an extended statistic. For this we note that the second order derivative of p_θ^T evaluated at $\theta = 0$, can be obtained by replacing two occurrences of $p_{i,\theta}$ by $(q_i - p_i)$. In formula,

$$\left. \frac{d^2}{d\theta^2} \{p_\theta^T \Phi\} \right|_{\theta=0} = \sum_{j=1}^{N(T)} \sum_{k=j+1}^{N(T)+1} \left(\prod_{i=k+1}^{N(T)+1} p_i(q_i - p_i) \prod_{i=j+1}^{k-1} p_i(q_i - p_i) \prod_{i=1}^{j-1} p_i \right) \Phi. \quad (18)$$

Adjusting for variance, we take as test statistic the second order Taylor series of p_θ^T developed at $\theta = 0$ and evaluated at $\theta = 1$:

$$G_1 = \sqrt{n} \frac{\left. \frac{d}{d\theta} \{p_\theta^T \Phi\} \right|_{\theta=0} + \frac{1}{2} \left. \frac{d^2}{d\theta^2} \{p_\theta^T \Phi\} \right|_{\theta=0}}{\left(\left. \frac{d}{d\theta} \{p_\theta^T \Phi\} \right|_{\theta=0} + \frac{1}{2} \left. \frac{d^2}{d\theta^2} \{p_\theta^T \Phi\} \right|_{\theta=0} \right)}, \quad (19)$$

where $s(\left. \frac{d}{d\theta} \{p_\theta^T \Phi\} \right|_{\theta=0} + \frac{1}{2} \left. \frac{d^2}{d\theta^2} \{p_\theta^T \Phi\} \right|_{\theta=0})$ is the standard error of the simulated statistic in (19) (for simulating of this gradient we will use CRN). Due to its representational equivalence to Eq. (11), the random variable G_1 is also t-distributed with $n - 1$ degrees of freedom, i.e., $G_1 \sim t_{n-1}$. We can then directly write a hypothesis test as

$$\begin{aligned} \text{second-order test} \quad H_0: G_1 &= 0 \\ H_1: G_1 &\neq 0. \end{aligned} \quad (20)$$

If the accumulated effect of the first two derivatives is small, then inserting only a few jumps has no significant effect on the payoff (technically speaking we conjecture that the combined contribution of all higher-order derivatives in the Taylor series is small). If, on the other hand, the accumulated effect of the first two derivatives is significant, then it is reasonable to assume that $p^T \Phi$ is different from $q^T \Phi = p_{\theta=1}^T$. This indicates that the model with jumps is the appropriate model. This effect is captured in the power of the test: the probability $\mathbb{P}(|G_1| > t_{n-1, 1-\alpha/2})$, where $t_{n-1, 1-\alpha/2}$ is the $1 - \alpha/2$ quantile of the t_{n-1} distribution.

Remark 1. On the event $N(T) = 0$, there are no jumps occurring between 0 and T , and $G_1 = G_2 = 0$. However, comparing $Y(t)$ and $X(t)$ on this event will have a non-zero value as the value of the drift under the EMM of $Y(t)$ contains the jump rate, see, Example 1 and compare (4) with (6).

Remark 2. The validity of the use of the t-distribution follows from the Central Limit Theorem for independent and identically generated random variables, where we assume that the second moment of the estimator is finite. Sufficient conditions will be provided in the next section.

Remark 3. Our framework covers model risk assessment of American vanilla options evaluated by simulation as well. The most popular approach for evaluation an American vanilla option by simulation, the Longstaff–Schwartz algorithm Longstaff & Schwartz (2001), approximates the early exercise boundary at a finite number of early exercise times $t = (t_0, \dots, t_e)$, with $0 < t_1 < \dots < t_e := T$, analogous to the observation times for path-dependent European options. The price paths are simulated forward in time between successive early exercise times. The requirement of the strong Markov property is sufficient to simulate American options via the Longstaff–Schwartz algorithm. The approximation of the quantities involved in the backwards dynamic programming step via a least-squares approximation is unaffected by the specific choice of Markovian price path.

3. Taylor series representation

The basis of estimating the first and second-order derivatives is the Strong Markov property. Here the evolution of the probability distribution of Markov process after a stopping time τ is the same as if the process stops and restarts after a fixed time, i.e. for $0 < s, \tau < t$ $P^s P^{t-s} = P^\tau P^{t-\tau}$. The occurrences of jumps of a Poisson process are examples of stopping times. Between successive jump times, both asset pricing models are merely diffusions in which the Wiener process can be simulated by the same method. This is the basis of our estimation procedure.

Assumptions

- (A1) Stochastic processes X and Y have the strong Markov property.
 (A2) The contingent claim Φ has a growth rate that is no more than linear, i.e., for $x = (x_1, \dots, x_\eta)$, for some constant $K > 0$

$$|\Phi(x)| \leq K \left(1 + \sum_{j=1}^{\eta} |x_j| \right).$$

- (A3) Both stochastic processes are L^2 integrable, i.e., $\mathbb{E}[X^2(T)], \mathbb{E}[Y^2(T)] < \infty$.

Assumption (A2) characterizes the contingent claims we can consider when comparing price models.

For $l = (l_1, \dots, l_n) \in \{0, 1\}^n$, and set $|l| = \sum_{i=1}^n l_i$ the count of 1's in l . Moreover, for l with $|l| = m$, denote by i_k the position of the k th value of 1 in l , for $1 \leq k \leq m$. In case that l is the zero vector, we let $i_1 = n + 1$. For $l = (l_1, \dots, l_n) \in \{0, 1\}^n$ let

$$p(l) = \left(\prod_{i=i_m+1}^n p_i \right) (q_{i_m} - p_{i_m}) \left(\prod_{i=i_{m-1}+1}^{i_m-1} p_i \right) \cdots \left(\prod_{i=i_1+1}^{i_2-1} p_i \right) (q_{i_1} - p_{i_1}) \left(\prod_{i=1}^{i_1-1} p_i \right),$$

where we let $\prod_{i \in \emptyset} p_i = 1$.

Theorem 1. *Provided that assumptions (A1) to (A3) hold, we have conditioned on the jump sequence*

$$Q^T \Phi = \sum_{n=0}^{N(T)} \sum_{l \in \{0,1\}^n, |l|=n} P(l) \Phi. \quad (21)$$

Moreover,

$$\mathbb{E}_{N(T)}[Q^T \Phi] = \mathbb{E}_{N(T)} \left[\sum_{n=0}^{N(T)} \sum_{l \in \{0,1\}^n, |l|=n} P(l) \Phi \right],$$

where the expectation is taken with respect to the Poisson jump process.

Proof. For ease of exposition, we prove the statement for $N(T) = 1$. The actual statement then follows by induction. Given $N(T) = 1$, we have

$$p_\theta^T = p_{1,\theta} p_{2,\theta} = ((1 - \theta)p_1 + \theta q_1)((1 - \theta)p_2 + \theta q_2).$$

Differentiating with respect to θ yields

$$\frac{d}{d\theta} p_\theta^T = (p_1 - q_1)((1 - \theta)p_2 + \theta q_2) + ((1 - \theta)p_1 + \theta q_1)(q_2 - p_2)$$

and

$$\frac{d^2}{d\theta^2} p_\theta^T = 2(p_1 - q_1)(q_2 - p_2).$$

Moreover, all derivatives of order $n \geq N(T) + 1$ are zero. Hence, we can write p_θ^T as Taylor series developed in $\theta = 0$ through

$$p_\theta^T = p_0^T + \theta \frac{d}{d\theta} p_0^T + \frac{\theta^2}{2} \frac{d^2}{d\theta^2} p_0^T,$$

which establishes the first part of the theorem.

We now turn to the second part. Since Φ does not grow faster than linear, we may bound Φ by the sum of the process values at observation points and the jumps. Independent of l we thus have

$$\mathbb{E}[|P(l)\Phi|] \leq K \mathbb{E} \left[\sum_{j=0}^{\eta} X_j \right] + K \mathbb{E}[N(T)] \mathbb{E}[J_1] < \infty,$$

where (A2) implies the first sum on the right-hand side is finite. The derivatives grow combinatorially. More specifically, the n th order derivative has as many $p(l)$ elements as there are possibilities to place n times a "one" on $N(T) + 1$ places, which is bounded by $(N(T) + 1)^n$. Since $N(T)$ has a finite moment generating function, we conclude that taking expectation with respect to $N(T)$ is justified. $\square$

4. Numerical study

We will demonstrate the performance of our procedure based on two competing pairs of models. In [Example 1](#) the performance is assessed between GBM and Merton jump-diffusion models in the vanilla call setting. A deeper assessment is provided in [Example 2](#) between the Mean Reversion (MR) and the Mean Reverting Jump Diffusion (MRJD) models in the vanilla put and Asian option setting. These types of models are the most widely used models for equities and FX (the first pair) and for interest rates and commodities (the second pair). To illustrate our statistical testing procedure, we will test for differences between option prices in the presence of jumps in multiple contexts and compare our results to the t-Test. We note that we do not compare our test with non-parametric tests (see, e.g., [Ait-Sahalia & Jacod \(2009\)](#); [Corsi, Pirino, & Renó \(2010\)](#); [Jacod, Glupeerberg, & Muller \(2017\)](#); [Lee & Mykland \(2008\)](#)) that tests on the underlying price process to statistically signify whether the price process has jumps. A numerical comparison between such non-parametric tests on the presence of jumps and our tests on the significance of jumps is further research.

Model Pair 1: H_0 : GBM Model vs. H_1 : Merton Model

For this model pair, we will examine a European call ATM option. With maturity price $S(T)$ and exercise price K , it has a well-known payoff

$$\text{Example 1: } \phi_{v,call}(S(T)) = \max\{S(T) - K, 0\}.$$

The baseline parameterization that we use is taken from [Table 4](#) in [Jorion \(1988\)](#), from the estimated parameters of the Merton Model of weekly data from a weighted index of stocks between January 1974 to December 1985. Both the contract and model parameter values are given in [Table 1](#). In particular, the parameterization for the GBM model is the diffusion component of the Merton model.

Model Pair 2: H_0 : Vasicek Model vs. H_1 : MRJD Model

The jump amplitude that we consider for the MRJD model is the asymmetric double exponential distribution, following the approach in [Kou \(2002\)](#). Let β^-, β^+ be the mean jump size in the positive and negative directions, respectively:

$$f_j(x) = \frac{1}{2} \left(\frac{1}{\beta^-} e^{-\frac{|x|}{\beta^-}} \mathbf{1}\{x < 0\} + \frac{1}{\beta^+} e^{-\frac{x}{\beta^+}} \mathbf{1}\{x \geq 0\} \right).$$

For this model pair, we will examine three call/put option payoffs: the vanilla put option, the discrete arithmetic Asian call option, averaging over four values, as well as an Asian put option, averaging over twelve times. This corresponds to three and eleven observation times, respectively, and these times are equidistant. The observation times are defined by $t_i = iT/o$. In formula,

$$\text{Example 2: } \Phi_{v,put}(S(T)) = \max\{K - S(T), 0\},$$

$$\text{Example 3: } \Phi_{a,call}(S(T)) = \max \left\{ \frac{1}{o} \sum_{i=1}^o S_i - K, 0 \right\},$$

$$\text{Example 4: } \Phi_{a,put}(S(T)) = \max \left\{ K - \frac{1}{o} \sum_{i=1}^o S_i, 0 \right\}.$$

For this model, we consider when the strike price is in-the-money for the put and out-of the money for the call. We utilize the parameters from the MRJD process from [Borovkova & Schmeck \(2017\)](#). These parameters were estimated from daily German EEX electricity spot prices from 12/03/2009 to 31/12/2013. We denote by $1/\beta := 1/\beta^+ + 1/\beta^-$ the harmonic sum of the mean jump sizes.

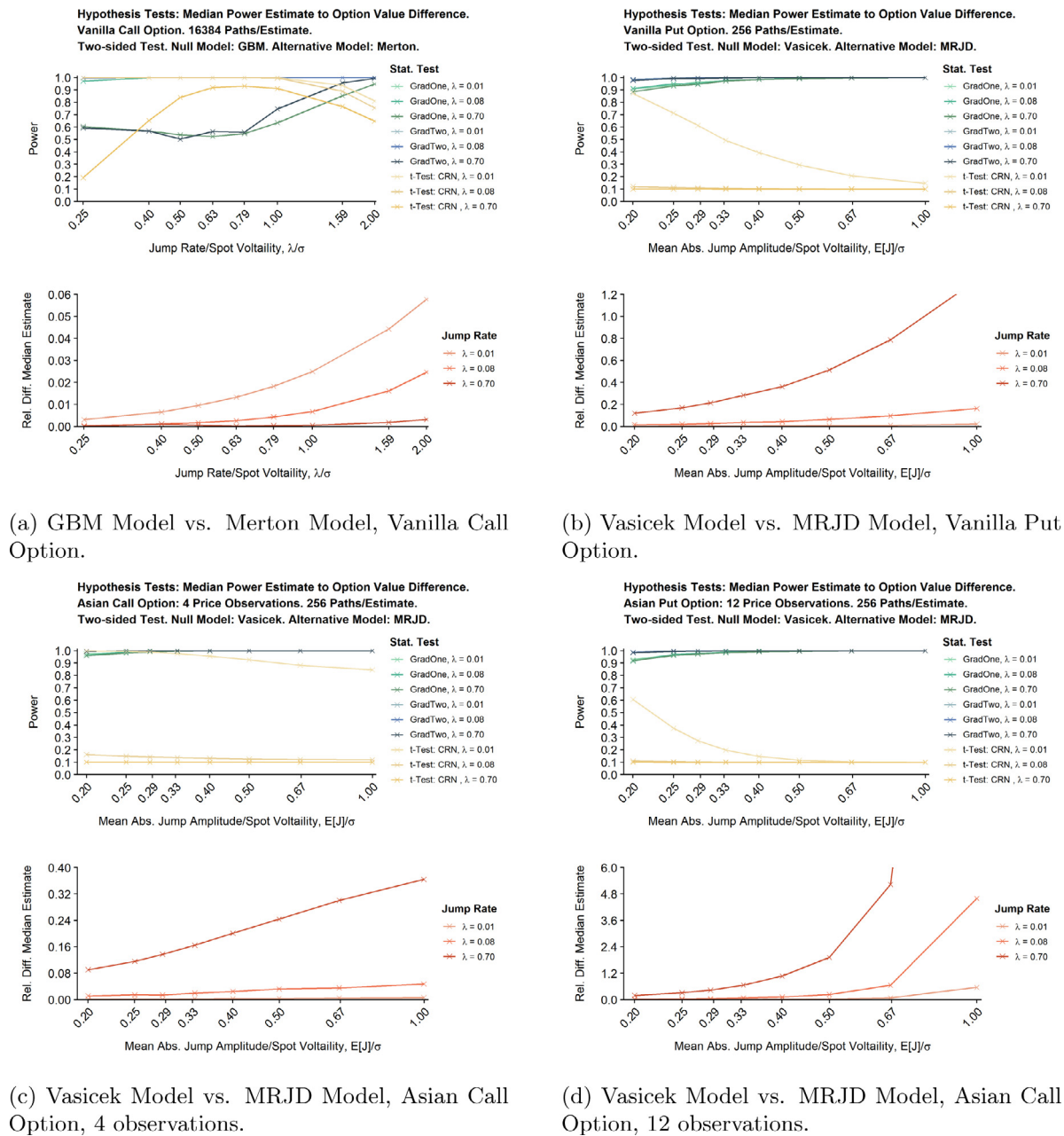


Fig. 1. Median of the power probability and the relative difference in median estimate for various model pairs and parameters of observation. The deviation for the alternative validation via the t-Test is set to $\delta = 0.2\%$ of the diffusion model option price.

Both the contract and model parameter values are given in Table 2. The parameterization for the Vasicek model is similarly the diffusion component of the MRJD model.

Other values for the risk-free rate were not considered. Because the test statistics are standardized quantities, a change in the interest rate, especially in the range of yields that short-term sovereign bonds trading in markets, would not substantially affect the statistical tests.

In all call/put option combinations, we vary the jump rate λ to ascertain the extent of influence this parameter has on the difference in option prices. In both model pairs, the jump rate is systematically modified in concert with a second parameter. For Model Pair 1, the spot volatility is modified. For Model Pair 2, the mean jump size in both directions (β^+ , β^-) is modified such that both

the mean jump size and the probability of a jump occurring in either direction is unchanged. For specific values of λ , for Model Pair 1, the parameter values are set according to the analysis ratio $= \lambda/\sigma$. For Model Pair 2, the parameters are set according to the analysis ratio $= \lambda/(\beta^+ - \beta^-)$. In our results, we primarily center on values of $\lambda = \{0.01, 0.08, 0.70\}$.

The results are presented when the option price is the mean of $n = 2^8 = 256$ paths. This is a choice between providing a strong interpretation of results and a parsimonious simulation size. Nonetheless, we do present results for values of n increasing by a factor of two. For Model Pair 1, we compute the option price for both the GBM and Merton models over $N = 500$ runs. For Model Pair 2, the option price is computed for both the Vasicek and MRJD models over $N = 300$ runs, where the number of runs is chosen

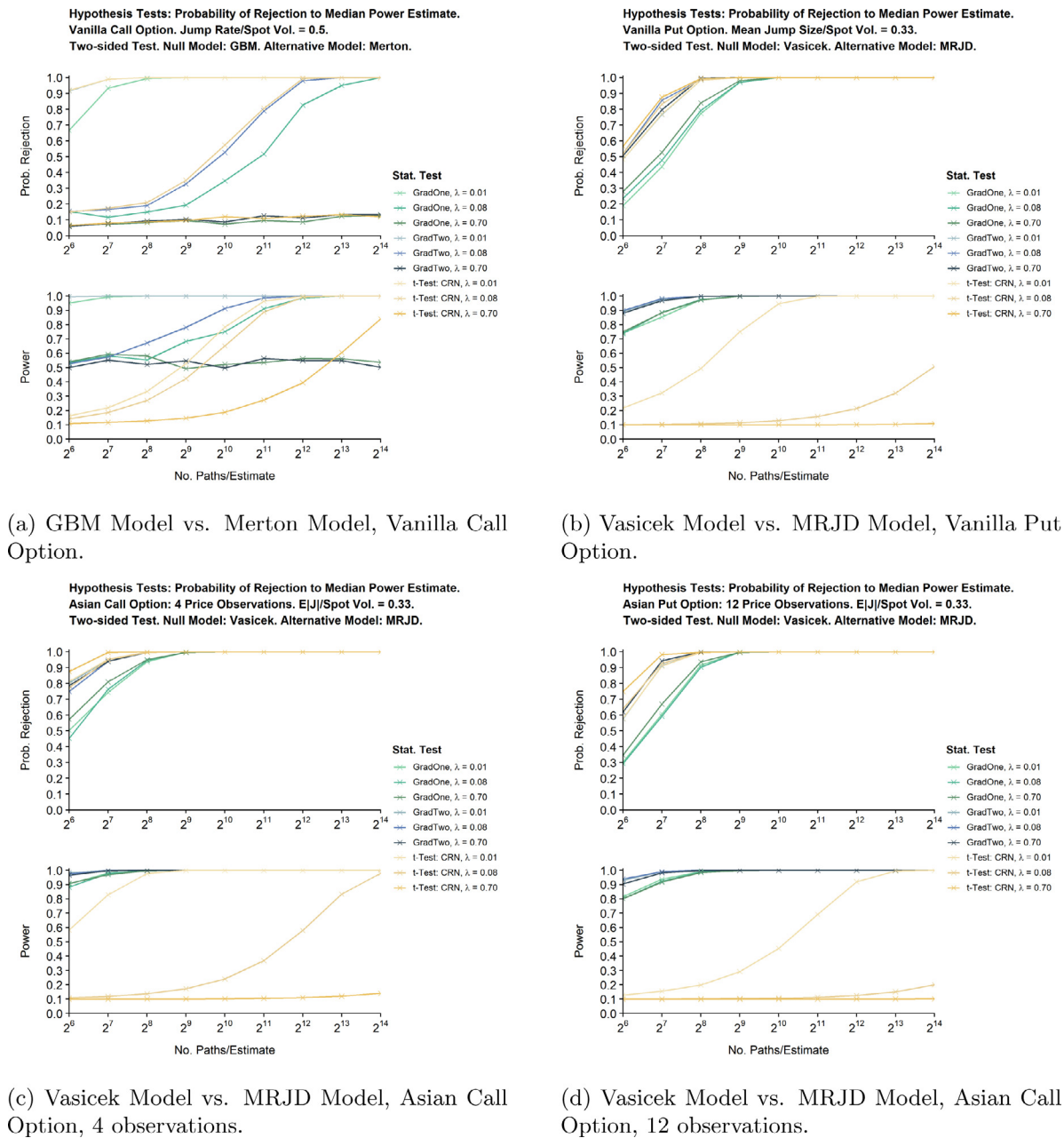


Fig. 2. Contrasting median probability of rejection and median power probability measures due to change of the number of simulated paths.

such that the running time of the numerical experiments remains feasible.

For each of the statistical validation approaches, a two-sided test is utilized with Type I-error of $\alpha = 0.10$. All implementations have been verified to be correct when compared to analytic results. For each of the statistical estimation approaches, antithetic sampling is used in the diffusion component as well as in the jump times. In addition, the probability of a jump occurring is also stratified.

4.1. Power of test statistics

We evaluate the performance of the statistical estimators, by (i) the probability of rejection, denoted by Prob. Rej., (ii) the median and estimate of the power probability denoted by

$q_{0.5}$ Power, and (iii) the relative difference in median and estimate. The relative difference in median estimate is measured by Rel. Diff. Median Estimate, which is measured by $|\widehat{P}\Phi_{0.5} - \widehat{Q}\Phi_{0.5}|/\widehat{P}\Phi_{0.5}$, where $\widehat{P}\Phi_{0.5}$, $\widehat{Q}\Phi_{0.5}$ are the estimated median option price for the respectively diffusion and jump-diffusion models. The equivalent relative mean estimate is measured by the same approach, except that median estimates are replaced by the mean of all estimates. Estimation by both the median and mean option prices shows that the findings of the numerical analysis is robust to measurement.

The probability of rejection estimates the proportion of runs in which the null hypothesis is rejected, stating that there is sufficient evidence to accept that the jump-diffusion model is a better representation. The power probability estimates the probability that the test statistic under the alterna-

tive hypothesis is outside the acceptance-rejection under the null hypothesis.

For the t-Test we consider option price deviations, see (12), of 0.1%, 0.2%, and 0.5% of the diffusion option price.

The first set of numerical results, measuring the median of the power probability and the relative difference in median estimate is presented in Fig. 1. For comparison, we provide the power probability and the relative difference in estimate measured in terms of mean values in Fig. 3 in Appendix A. The results as measured in terms of mean estimates exhibit identical behaviour like the one measured in terms of the median.

From Fig. 1, we can observe whether each of the gradient approaches is effective in computing the power probability. As the spot volatility decreases, the power probability increases towards one and can do in Examples 2–4 for low jump rate of large jump sizes, when the spot volatility is greater than the mean jump amplitude, and for estimate sizes of $2^8 - 2^{10}$. This is the anticipated behavior because the jump amplitude becomes relatively more pronounced than the movement in price due to volatility.

We observe a methodological problem with the t-test approach. If the spot volatility decreases whilst all other parameters remain fixed (notably, the one-sided jump values $\mathbb{E}[J^\pm]$ remain fixed), the influence of jumps must increase and normatively be present in a validation approach. This is, however, not detected by the t-test. This is exhibited by the difference in option price $d\hat{P}\Phi$, a proportion of the diffusion option price, because of this effect. The standard deviation measures the difference in option price, and if the jump amplitude becomes more prominent, the difference is more significant, and therefore the standard deviation is higher. Hence the change in the test statistic in (12) is less.

The same effect occurs if we use other choices for the difference in option price in (12). If a constant value is chosen, for instance, d , the same problem, albeit more mildly, will occur. As σ decreases, the denominator in (12) increases whilst the numerator remains constant. If the difference in option price is determined via $d\hat{Q}\Phi$, rejecting the appropriateness diffusion price model, a decrease in σ , the option price also decreases. For instance, the analytic solution of the Merton model call option, see Merton (1976), is a superposition of Black-Scholes call options predicated on the number of jumps observed. The power probability will increase for the t-Test approaches, choosing $d\hat{Q}\Phi$ as its nominated difference when the mean jump amplitude increases for a call option/ decreases for a put option.

From all model examples in Fig. 2, the probability of rejecting the null hypothesis occurs via the t-Test approach via the same or fewer sample paths for the gradient approach. The variance reduction in the *GradTwo* gradient estimation approach assists in rejecting the null hypothesis in fewer paths than in the *GradOne* estimation approach. As the t-Test approach gives incorrect estimates for the power probability, the gradient estimation approach is preferred.

The quality of outcome to ascertain model risk via the gradient-based statistical test is dependent on the quality and stability of the estimating procedure as well as the variability of the incoming data. As a change in hypothesis is caused by a change in the critical value pertaining to the used statistic, the hypothesis is less frequent than the variability of parameter outputs. For instance, the gradient-based test could clearly continue to reject the hypothesis test. In this hypothetical example, it is apparent that in-

cluding jumps would significantly improve the estimation accuracy, even though there are changes in parameter values due to automated estimation of parameters using data from a rolling time window.

4.2. Summary

The previous sections show the utility of the gradient method in discriminating between a diffusion and a jump-diffusion model. The gradient methods, particularly *GradTwo*, have a tendency to reject the diffusion model almost in proportion to the t-test validation method. However, the gradient methods can correctly detect the presence of jump-diffusion models and can increasingly do so when the parameter indicates an increased difference in the option price.

5. Concluding remarks

We presented a new statistical procedure to test whether an option pricing model that contains jumps in the underlying is significantly different (in terms of the resulting option prices) than a simpler pure diffusion model. We have demonstrated that our test, based on the technique of gradient estimation, is a significant improvement with respect to the simple t-test and is particularly powerful in situations when either the jump or the diffusion component is particularly dominant.

Our suggested statistical testing procedure is quite robust to errors in estimated parameters (and more robust than option pricing models for jump-diffusions, which are quite sensitive to parameter values). Our test is particularly robust to situations when jump frequency and/or amplitude is overestimated compared to the true values since for our test to resolve in favor of jumps, their frequency/amplitude needs to have some minimum value, which depends on the volatility of the diffusion component. It is well known that most jump calibration procedures suffer from exactly such overestimation, which is less damaging for our test. So our method offers a practical solution: first, perform a rough calibration of jump parameters, then apply our gradient-based test and, if it shows that jumps will have a significant effect on option prices, spend more time and effort on the better calibration of model parameters.

Our testing procedure can be implemented quite generically. The types of options for which the test should be performed and the types of models with/without jumps can be specified completely separately from each other and the test is then "built" from the existing building blocks for each particular situation. This is because path simulation is implemented completely separately from the evaluation of options for which the test should be performed, and if more or different options are added, then an additional option valuation module can be implemented at minimum computational cost and implementation effort. We are currently undertaking such an open-source implementation, to provide financial institutions with a free tool to test for significance of jumps on their option prices, and to assist them in this way in their model risk mitigation.

An interesting conceptual extension of our work would be to apply the mixture model for gradually mixing a base model with a jump model to Malliavin calculus. Also, an empirical analysis on the stability statistical test behaviour under different time scales and asset classes of market data.

Appendix A. Mean power to relative mean value plots

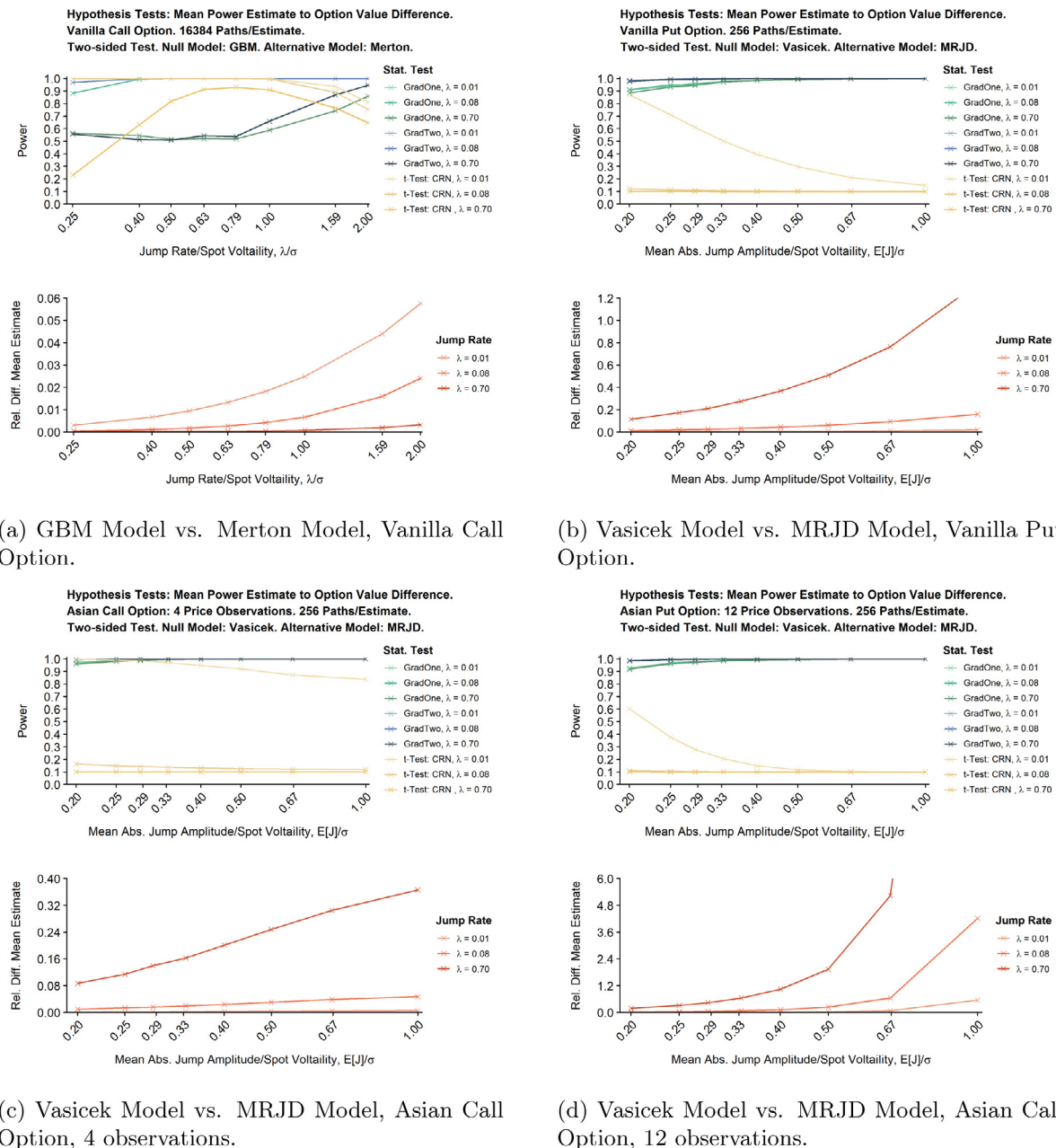


Fig. 3. Mean of the estimated power and the relative difference in mean estimate for various model pairs and varying observation parameters. The deviation for the alternative validation via the t-Test is set to $\delta = 0.2$ of the diffusion model option price.

References

- Ait-Sahalia, Y. (2004). Disentangling diffusion from jumps. *Journal of Financial Economics*, 74, 487–528.
- Ait-Sahalia, Y., & Jacod, J. (2009). Testing for jumps in a discretely observed process. *The Annals of Statistics*, 37, 184–222.
- Andersen, L., & Andreasen, J. (2000). Jump-diffusion processes: Volatility smile fitting and numerical methods for option pricing. *Review of Derivatives Research*, 4, 231–262.
- Asmussen, S., & Glynn, P. (2007). *Stochastic simulation: Algorithms and analysis*. New York, NY: Springer.
- Ball, C. A., & Torous, W. N. (1985). On jumps in common stock prices and their impact on call option pricing. *The Journal of Finance*, 40, 155–173.
- Bates, D. S. (1996). Jumps and stochastic volatility: Exchange rate processes implicit in deutsche mark options. *The Review of Financial Studies*, 9, 69–107.
- Benth, F. E., & Schmeck, M. D. (2014). Pricing and hedging options in energy markets using black-76. *The Journal of Energy Markets*, 7, 35–69.
- Borovkova, S., & Schmeck, M. D. (2017). Electricity price modeling with stochastic time change. *Energy Economics*, 63, 51–65.
- Capriotti, L., & Giles, M. B. (2011). Algorithmic differentiation: Adjoint Greeks made easy. Available at SSRN: <http://ssrn.com/abstract=1801522>.
- Cartea, A., & Figueroa, M. G. (2005). Pricing in electricity markets: A mean reverting jump diffusion model with seasonality. *Applied Mathematical Finance*, 12, 313–335.
- Clewlow, L., & Strickland, C. (2003). *Energy derivatives: Pricing and risk management*. Houston, TX: Lacima Publications.
- Cont, R. (2006). Model uncertainty and its impact on the pricing of derivative instruments. *Mathematical Finance*, 16, 519–547.
- Cont, R., & Hamida, S. b. (2005). Recovering volatility from option prices by evolutionary optimization. *Journal of Computational Finance*, 8, 43–76.

- Cont, R., & Tankov, P. (2004a). *Financial modelling with jump processes*. Boca Raton, FL: Chapman & Hall/CRC.
- Cont, R., & Tankov, P. (2004b). Nonparametric calibration of jump-diffusion option pricing models. *The Journal of Computational Finance*, 7, 1–49.
- Cont, R., & Tankov, P. (2006). Retrieving Lévy processes from option prices: Regularization of an ill-posed inverse problem. *SIAM Journal of Control Optimization*, 45, 1–25.
- Corsi, F., Pirino, D., & Renó, R. (2010). Threshold bipower variation and the impact of jumps on volatility forecasting. *Journal of Econometrics*, 159, 276–288.
- Derman, E. (1996). Model risk. *Risk*, 9, 34–37. 139–145
- Detering, N., & Packham, N. (2016). Model risk of contingent claims. *Quantitative Finance*, 16, 1357–1374.
- Dowdy, S., Weardon, S., & Chilko, D. (2005). *Statistics for research*. Hoboken, NJ: John Wiley & Sons.
- Erlwein, C., Benth, F. E., & Mamon, R. (2010). HMM filtering and parameter estimation of an electricity spot price model. *Energy Economics*, 32, 1034–1043.
- Guide for the Targeted Review of Internal Models (TRIM), 2017. European Central Bank: Banking Supervision"; available at https://www.bankingsupervision.europa.eu/ecb/pub/pdf/trim_guide.en.pdf.
- Giles, M. B. (2008). Multilevel monte carlo path simulation. *Operations Research*, 56, 607–617.
- Glasserman, P., & Xu, X. (2014). Robust risk measurement and model risk. *Quantitative Finance*, 14, 29–58.
- Heidergott, B., & Volk-Makarewicz, W. (2015). A measure-valued differentiation approach to sensitivities of quantiles. *Mathematics of Operations Research*, 41, 293–317.
- Hilliard, J. E., & Reis, J. A. (1999). Jump processes in commodity futures and options pricing. *American Journal of Agricultural Economics*, 81, 273–286.
- Jacod, J., Glupeerberg, C., & Muller, G. (2017). Testing for non-correlation between price and volatility jumps. *Journal of Econometrics*, 197, 284–297.
- Jorion, P. (1988). On jump processes in the foreign exchange and stock markets. *The Review of Financial Studies*, 1, 427–445.
- Kou, S. G. (2002). A jump-diffusion model for option pricing. *Management Science*, 48, 1086–1101.
- Larsson, K., & Nossman, M. (2011). Jumps and stochastic volatility in oil prices: Time series evidence. *Energy Economics*, 33, 504–514.
- Lee, S. S., & Mykland, P. A. (2008). Jumps in financial markets: A new nonparametric test and jump dynamics. *The Review of Financial Studies*, 21, 2535–2563.
- Longstaff, F. A., & Schwartz, E. S. (2001). Valuing american options by simulation: A simple least-squares approach. *The Review of Financial Studies*, 14, 113–147.
- Mayer, K., Schmid, T., & Weber, F. (2015). Modeling electricity spot prices: Combining mean reversion, spikes, and stochastic volatility. *The European Journal of Finance*, 21, 292–315.
- Merton, R. C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3, 125–144.
- Morini, M. (2011). *Understanding and managing model risk: A practical guide for quants, traders and validators*. Chichester, West Sussex: John Wiley & Sons.
- Pflug, G. (1996). *Optimization of stochastic models: The interface between simulation and optimization*. Norwell, MA: Kluwer.
- Rogers, L. C. G., & Williams, D. (2000). *Diffusions, Markov processes and martingales: Volume 1, foundations*. Cambridge, Cambridgeshire: Cambridge University Press.